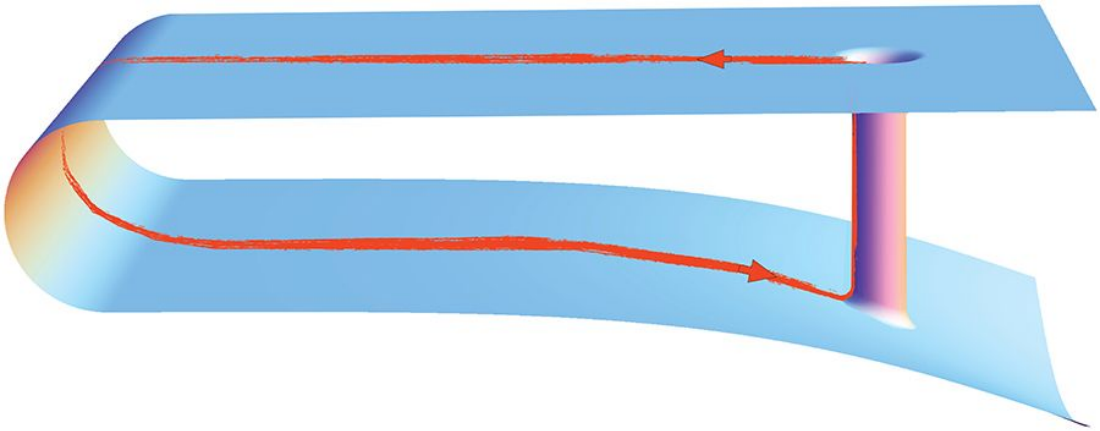


Fig. 30.4. Quantum fluctuations of light, traveling along the red path, build up into a crescendo explosion at the moment the wormhole is becoming a time machine.

Many researchers have struggled, over the past twenty years, to prove or disprove Hawking's chronology protection conjecture. The bottom line today, I think, remains the same as in the early 1990s, when he and I were debating the issue: Only the laws of quantum gravity know for sure.

Time Travel with a Bulk

All this research and conclusions—educated guesses—are based on the laws of physics that prevail *if there is no bulk with a large fifth dimension*. What happens to time travel if a large bulk *does* exist, as in *Interstellar*?

We physicists find Einstein's relativistic laws so compelling that we suspect they hold in the bulk as well as in our brane. So Lisa Randall, Raman Sundrum, and others have extended his laws into the five-dimensional bulk by one simple step: adding a new dimension to space. That extension proceeds mathematically in a straightforward and beautiful manner, which makes us physicists think we may be on the right track. In my interpretation of the movie, Professor Brand uses this